Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\boldsymbol{x}_i\}_{i=1}^n$ where $\boldsymbol{x}_i \in \mathbb{R}^p$ for all $i = 1, \ldots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\boldsymbol{x}_i$'s as realizations of independent random variables $\boldsymbol{X}_i = \boldsymbol{W}\boldsymbol{Z}_i + \boldsymbol{\epsilon_i}$, where $\boldsymbol{W}$ is an unknown $p \times q$ loading matrix of full rank, $\boldsymbol{Z}_i \sim N_q(\mathbf{0}, \boldsymbol{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon_i} \sim N_p(\mathbf{0}, \sigma^2\boldsymbol{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n, \boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\boldsymbol{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ. Θ is thus the collection of all tuples $(\boldsymbol{W}, \sigma^2)$ such that $\boldsymbol{W}$ is a real $p \times q$ matrix with rank$(\boldsymbol{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \left\{N_p\left(\mathbf{0}, \boldsymbol{W}_\theta\boldsymbol{W}_\theta^T + \sigma_\theta^2\boldsymbol{I}_p\right)\right\}_{\theta\in\Theta}. \tag{1.1}$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\boldsymbol{A}, s) \in \Theta$ and $(\boldsymbol{B}, t) \in \Theta$, we define $(\boldsymbol{A}, s) \sim (\boldsymbol{B}, t)$ to mean that they give rise to the same covariance matrix,namely when $\boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p = \boldsymbol{B}\boldsymbol{B}^T + t\boldsymbol{I}_p$. The equivalence classes are denoted by $[(\boldsymbol{A}, s)]$ or more briefly $[\boldsymbol{A}, s]$. Unless otherwise mentioned, $(\boldsymbol{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\boldsymbol{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\boldsymbol{A}, s] = \{(\boldsymbol{A}\boldsymbol{Q}, s) \mid \boldsymbol{Q} \in O(q)\}$. Therefore, $M := \Theta/\sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and the map $\pi : \Theta \to M$ defined by

$$\pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s] \tag{1.2}$$

is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomertic with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28), meaning that

$$g^{\Theta}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right) = g_{[\boldsymbol{A},s]}\left(d\pi_{(A,s)}((\boldsymbol{V}, c)), d\pi_{(A,s)}((\boldsymbol{U}, b))\right), \tag{1.3}$$

where $g^{\Theta}_{(\boldsymbol{A},s)}$ is the standard Euclidean metric given by

$$g^{\Theta}_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{U}, b)\right) = \operatorname{tr}\left(\boldsymbol{V}\boldsymbol{U}^T\right) + cb, \tag{1.4}$$

whenever $(\boldsymbol{A}, s) \in \Theta$, $(\boldsymbol{V}, c)$, $(\boldsymbol{U}, b) \in T_{(\boldsymbol{A},s)}\Theta \cong \mathbb{R}^{p\times q} \times \mathbb{R}$. For later convenience, let us define two functions $\Phi : \Theta \to \mathbb{R}^{p\times p}$ and $\boldsymbol{\Sigma} : M \to \mathbb{R}^{p\times p}$ by setting, for every $[\boldsymbol{A}, s] \in M$:

$$\boldsymbol{\Sigma}([\boldsymbol{A}, s]) := \Phi((\boldsymbol{A}, s)) := \boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p. \tag{1.5}$$

In other words, $\boldsymbol{\Sigma} \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))\}_{[\boldsymbol{A},s]\in M} \tag{1.6}$$

and this is an identifiable model we shall work with.

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces.

Proposition 1.

1. The map $\boldsymbol{\Sigma}$ defined in (1.5) is a smooth injective immersion.

2. At every $(\boldsymbol{A}, s) \in \Theta$, the verticle tangent space

$$V_{(\boldsymbol{A},s)} := \ker(d\pi_{(\boldsymbol{A},s)}) = \ker(d\Phi_{(\boldsymbol{A},s)}) \tag{1.7}$$

   equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

3. At every $(\boldsymbol{A}, s) \in \Theta$, the horizontal tangent space $H_{(\boldsymbol{A},s)} := V^{\perp}_{(\boldsymbol{A},s)}$ equals to the collection of all tangent vectors of the form $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $q \times q$ matrices $\boldsymbol{V}$ satisfying $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$.

Proof. Fix a point $(\boldsymbol{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(d\Phi_{(\boldsymbol{A},s)})$, where Φ is given in (1.5). For $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ by setting $\gamma(t) = (\boldsymbol{A} + t\boldsymbol{V}, s + tc)$. Then:

$$d\Phi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c)) = \left.\frac{d}{dt}\right|_{t=0} \Phi(\gamma(t)) = \boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p.$$

If $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$, then

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p = \boldsymbol{0}. \tag{1.8}$$

If $\boldsymbol{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\boldsymbol{A}$, then $\boldsymbol{A}^T\boldsymbol{u} = \boldsymbol{0}$ and

$$0 = \boldsymbol{u}^T(\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p)\boldsymbol{u} = c\,\boldsymbol{u}^T\boldsymbol{u}.$$

Given that $\boldsymbol{u} \neq \boldsymbol{0}$, we must have $c = 0$ and (1.8) reduces to

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T = \boldsymbol{0}. \tag{1.9}$$

Apply $\boldsymbol{u}$ to the right to get:

$$\boldsymbol{A}\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Given that $\boldsymbol{A}$ has full rank, rank nullity theorem forces

$$\boldsymbol{V}^T\boldsymbol{u} = \boldsymbol{0}.$$

Therefore, we have proven that every vector $\boldsymbol{u}$ orthgonal to columns vectors of $\boldsymbol{A}$ will also be orthgonal to column vectors of $\boldsymbol{V}$. This means $\boldsymbol{V} = \boldsymbol{A\Omega}$ for some $q \times q$ matrix $\boldsymbol{\Omega}$. Substitute this back to (1.9) gives

$$\boldsymbol{A}(\boldsymbol{\Omega}^T + \boldsymbol{\Omega})\boldsymbol{A}^T = \boldsymbol{0}.$$

Applying $(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T$ to the left and $\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A})^{-1}$ to the right yields $\boldsymbol{\Omega}^T + \boldsymbol{\Omega} = \boldsymbol{0}$. Therefore, $\boldsymbol{\Omega}$ is skew-symmetric. Conversely, (1.9) is satisfied by $\boldsymbol{V} = \boldsymbol{A\Omega}$ with any skew-symmetric matrix $\boldsymbol{\Omega}$. We have proven that $\ker(d\Phi_{(\boldsymbol{A},s)})$ equals to the collection of all tangent vectors of the form $(\boldsymbol{A\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.7). Clearly, $\boldsymbol{\Sigma}$ is well-defined and is injective. The smoothness of $\boldsymbol{\Sigma}$ is guaranteed by relation $\boldsymbol{\Sigma} \circ \pi = \Phi$ (see for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\boldsymbol{A},s)} = d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]} \circ d\pi_{(\boldsymbol{A},s)}. \tag{1.10}$$

It follows immediately that $\ker(d\Phi_{(\boldsymbol{A},s)}) \subseteq \ker(d\pi_{(\boldsymbol{A},s)})$. The other direction will follow if we can prove that $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective, or equivalently that $\boldsymbol{\Sigma}$ is an immersion.

To do so, suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ is such that $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = 0$. We prove that $\boldsymbol{v} = \boldsymbol{0}$. Let $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ be the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$, meaning that $(\boldsymbol{V}, c)$ is the unique horizontal tangent vector at $(\boldsymbol{A}, s)$ such that $\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$. (1.10) then implies that $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\boldsymbol{V} = \boldsymbol{A\Omega}$ for some skew-symmetric $\boldsymbol{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ defined by $\gamma(t) = (\boldsymbol{A}e^{t\boldsymbol{\Omega}}, s)$. Clearly $\gamma(0) = (\boldsymbol{A}, s)$ and $\gamma'(0) = (\boldsymbol{A\Omega}, 0)$. Moreover, the matrix $e^{t\boldsymbol{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\boldsymbol{\Omega}$ so that $\pi(\gamma(t)) = [\boldsymbol{A}, s]$ for all t. We have

$$\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{A\Omega}, 0)) = d\pi_{(\boldsymbol{A},s)}(\gamma'(0)) = \frac{d}{dt}\bigg|_{t=0} \pi(\gamma(t)) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M.$$

Therefore, $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective and the verticle tangent space is the same as $\ker(d\Phi_{(\boldsymbol{A},s)})$.

To compute the horizontal tangent space, any $(\boldsymbol{V}, c) \in V^{\perp}_{(\boldsymbol{A},s)}$ will satisfy

$$0 = g_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{A\Omega}, 0)\right) = \operatorname{tr}\left(\boldsymbol{A}^T\boldsymbol{V}\boldsymbol{\Omega}\right) \tag{1.11}$$

for all skew-symmetric $\boldsymbol{\Omega}$. Decomposing $\boldsymbol{A}^T\boldsymbol{V} = \boldsymbol{S} + \boldsymbol{K}$, where $\boldsymbol{S}$ is the symmetric part and $\boldsymbol{K}$ is the skew-symmetric part, we find out that

$$0 = \operatorname{tr}(\boldsymbol{S\Omega}) + \operatorname{tr}(\boldsymbol{K\Omega}) = 0 + \operatorname{tr}(\boldsymbol{K\Omega}) = \operatorname{tr}(\boldsymbol{K\Omega}).$$

Setting $\boldsymbol{\Omega} = \boldsymbol{K}^T$ will yield that $\boldsymbol{K} = \boldsymbol{0}$. Therefore, $\boldsymbol{A}^T\boldsymbol{V}$ is a symmetric matrix. Conversely, any $\boldsymbol{V}$ such that $\boldsymbol{A}^T\boldsymbol{V}$ is symmetric will satisfy (1.11). The proof is complete. □

Proposition 2. Given $[\boldsymbol{A}, s] \in M$ and $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$, geodesics in M starting at $[\boldsymbol{A}, s]$ with initial velocity $\boldsymbol{v}$ takes the format of curves $\gamma : I \to M$, where I is an open interval containing 0, defined by $\gamma(t) = [\boldsymbol{A} + t\boldsymbol{V}, s + tc]$, where $(\boldsymbol{V}, c) \in H_{(\boldsymbol{A},s)}$ is the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$. Consequently, the exponential map is computed as $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A} + \boldsymbol{V}, s + c]$, at least well-defined over an open ball $\mathcal{E}_{[\boldsymbol{A},s]}$ of $\boldsymbol{0} \in T_{[\boldsymbol{A},s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$ of radius $\min(\sigma_{\min}(\boldsymbol{A}), s)$, where $\sigma_{\min}(\boldsymbol{A})$ denotes the smallest singular value of $\boldsymbol{A}$.

all $\boldsymbol{v}$ such that $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v},\boldsymbol{v})} = |\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$, where $\sigma_{\min}(\boldsymbol{A})$ is the smallest singular value of $\boldsymbol{A}$.

Proof. First of all, let us quickly check that the curve γ is well defined. Indeed, if $(\boldsymbol{AQ}, s)$, where $\boldsymbol{Q} \in O(q)$, is another representative of $[\boldsymbol{A}, s]$, let $(\boldsymbol{V}', c')$ be the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{AQ}, s)$. With the function $R_{\boldsymbol{Q}} : \Theta \to \Theta$ defined by $R_{\boldsymbol{Q}}((\boldsymbol{A}, s)) = (\boldsymbol{AQ}, s)$, we have

$$\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d(\pi \circ R_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d\pi_{(\boldsymbol{AQ},s)}(dR_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d\pi_{(\boldsymbol{AQ},s)}(\boldsymbol{VQ}, c).$$

Thus, $\boldsymbol{V}' = \boldsymbol{VQ}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve.

A well-known property of geodesics in case of Riemannian submersion (see for example [O'Neill(1967)] Section 2 Corollary 2 or [Valencia(2021)] Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M. Therefore, $\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A}+\boldsymbol{V}, s+c]$ is well defined for all vectors $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ of small enough norm making sure that $(\boldsymbol{A}+\boldsymbol{V}, s+c) \in \Theta$. To have a quantified upper bound of the norm, we cite Weyl's perturbation inequality (Chapter 7.3 Problem 16 equation (7.3.15) of [Horn and Johnson(2013)]) to get

$$\sigma_{\text{min}}(\boldsymbol{A}+\boldsymbol{V}) \geqslant \sigma_{\text{min}}(\boldsymbol{A}) - \sigma_{\text{max}}(\boldsymbol{V}) \geqslant \sigma_{\text{min}}(\boldsymbol{A}) - |\boldsymbol{v}| > 0,$$

whenever $|\boldsymbol{v}| < \min(\sigma_{\text{min}}(\boldsymbol{A}), s)$. Here $\sigma_{\text{max}}(\boldsymbol{V})$ denotes the largest singular value of $\boldsymbol{V}$, who is smaller than the Euclidean norm of $\boldsymbol{V}$ and hence norm of $\boldsymbol{v}$. Thus, for all $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$, $\operatorname{rank}(\boldsymbol{A}+\boldsymbol{V}) = q$ and the exponential map is well-defined. $\square$

To compute the logarithmic map, we need first preparatory lemmas that work out an orthogonal matrix based on given points in M.

Lemma 3. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix and $\boldsymbol{Q} \in O(q)$ are such that $\boldsymbol{LQ}$ is symmetric, then there exists matrices $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$, a $q \times q$ diagonal matrix $\boldsymbol{D}$ whose diagonal entries are 1 or -1, and a $q \times q$ diagonal matrix $\boldsymbol{S}$ containing the singular values of $\boldsymbol{L}$ in increasing order such that $\boldsymbol{L} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$.

Proof. Let us start by applying the singular value decomposition of $\boldsymbol{L}$ to get matrices $\boldsymbol{R}' \in O(q)$, $\boldsymbol{P}' \in O(q)$ and $\boldsymbol{S}$ such that $\boldsymbol{L} = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T$. Substituting the given condition that $\boldsymbol{LQ}$ is symmetric will yield that $\boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$, where $\boldsymbol{J} := (\boldsymbol{P}')^T\boldsymbol{Q}\boldsymbol{R}'$.

Clearly $\boldsymbol{J} \in O(q)$. The equation

$$\boldsymbol{S}^2 = \boldsymbol{SJJ}^T\boldsymbol{S} = (\boldsymbol{SJ})(\boldsymbol{SJ})^T = (\boldsymbol{J}^T\boldsymbol{S})(\boldsymbol{J}^T\boldsymbol{S})^T = \boldsymbol{J}^T\boldsymbol{S}^2\boldsymbol{J}$$

shows that

$$\boldsymbol{JS}^2 = \boldsymbol{S}^2\boldsymbol{J}.$$

Looking at each entry of both size, we end up with

$$\boldsymbol{J}_{ij}(\boldsymbol{S}_{jj}^2 - \boldsymbol{S}_{ii}^2) = 0$$

for $i, j = 1, \ldots q$. Given that $\boldsymbol{L}$ is invertible, its singular values are all strictly positive. Therefore, in fact we have

$$\boldsymbol{J}_{ij}(\boldsymbol{S}_{jj} - \boldsymbol{S}_{ii}) = 0 \tag{1.12}$$

for all i and all j so that $\boldsymbol{JS} = \boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$. Invert $\boldsymbol{S}$ yields that $\boldsymbol{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\boldsymbol{J}^2 = \boldsymbol{I}_q$ and the eigenvalues of $\boldsymbol{J}$ must be 1 or -1. Another look at (1.12) shows that $\boldsymbol{J}$ is a block-diagonal matrix, diagonal if and only if all diagonal entries of $\boldsymbol{S}$ are distinct. Furthermore, each block of $\boldsymbol{J}$ must still be orthogonal and symmetric.

We now diagonalize $\boldsymbol{J}$ based on the above information. Suppose that $\boldsymbol{S} = \oplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{I}_{m_i}$, where $\boldsymbol{s}_1 > \ldots > \boldsymbol{s}_k$ are the unique diagonal entries of $\boldsymbol{S}$ and each repeats for m_i times. Thus, $\boldsymbol{J}$ has k blocks. Write the ith block by $\boldsymbol{J}_i$ so that $\boldsymbol{J} = \oplus_{i=1}^k \boldsymbol{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\boldsymbol{J}_i = \boldsymbol{K}_i\boldsymbol{D}_i\boldsymbol{K}_i^T$, where $\boldsymbol{K}_i \in O(m_i)$ and $\boldsymbol{D}_i$ is an $m_i \times m_i$ diagonal matrix with 1 or -1 diagonal entry. Put $\boldsymbol{K} = \oplus_{i=1}^k \boldsymbol{K}_i$ and $\boldsymbol{D} = \oplus_{i=1}^k \boldsymbol{D}_i$. Then $\boldsymbol{K} \in O(q)$, $\boldsymbol{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\boldsymbol{J} = \boldsymbol{KDK}^T$. This involved construction leads us to the outcome that $\boldsymbol{K}$ in fact commutes with $\boldsymbol{S}$:

$$\boldsymbol{KS} = \bigoplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{K}_i = \boldsymbol{SK}.$$

Put $\boldsymbol{P} = \boldsymbol{P}'\boldsymbol{K}$, $\boldsymbol{R} = \boldsymbol{R}'\boldsymbol{K}$. We have

$$\boldsymbol{Q} = \boldsymbol{P}'\boldsymbol{J}(\boldsymbol{R}')^T = \boldsymbol{P}'\boldsymbol{KDK}^T(\boldsymbol{R}')^T = \boldsymbol{PDR}^T$$

and

$$\boldsymbol{RSP}^T = \boldsymbol{R}'\boldsymbol{KSK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{SKK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T = \boldsymbol{L}.$$

This completes the proof. □

Lemma 4. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix with singular value decomposition $\boldsymbol{L} = \boldsymbol{RSP}^T$, then

$$\boldsymbol{PR}^T = (\boldsymbol{L}^T\boldsymbol{L})^{-\frac{1}{2}}\boldsymbol{L}^T.$$

This means, although neither $\boldsymbol{P}$ nor $\boldsymbol{R}$ is uniquely determined, $\boldsymbol{PR}^T$ is uniquely determined.

Proof. Given that

$$\boldsymbol{L}^T\boldsymbol{L} = \boldsymbol{PS}^2\boldsymbol{P}^T,$$

we have

$$(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}} = \boldsymbol{PSP}^T.$$

Therefore:

$$\boldsymbol{L} = \boldsymbol{RP}^T\boldsymbol{PSP}^T = \boldsymbol{RP}^T(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}.$$

The result follows from inverting $(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}$ and take transpose. □

Proposition 5. For every $[\boldsymbol{A}, s] \in M$ there exists an open neighborhood $U \subset M$ containing $[\boldsymbol{A}, s]$ such that

1. For every $[\boldsymbol{B}, t] \in U$, $\boldsymbol{A}^T\boldsymbol{B}$ is invertible.
2. The logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]} : U \to T_{[\boldsymbol{A},s]}M$ is well-defined and is smooth in U. In fact:

   $$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BQ}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}, t - s)),$$

   where

   $$\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) = (\boldsymbol{B}^T\boldsymbol{AA}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A} \in O(q).$$

Proof. First of all, if $\boldsymbol{R} \in O(q)$ and $\boldsymbol{P} \in O(q)$, then $\boldsymbol{Q}(\boldsymbol{AP}, \boldsymbol{BR}) = \boldsymbol{R}^T\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B})\boldsymbol{P}$ and the same computation as in the proof of Proposition 2 shows that the formula of logarithmic map is indeed well-defined.

Given that the determinant is a continuous function and $\det\big(\boldsymbol{A}^T\boldsymbol{A}\big) > 0$, there exists a neighborhood $U_\Theta \subset \Theta$ containing $(\boldsymbol{A}, s)$ such that $\det\big(\boldsymbol{A}^T\boldsymbol{B}\big) > 0$ whenever $(\boldsymbol{B}, t) \in U_\Theta$. Thus, for any $[\boldsymbol{B}, t] \in U_1 := \pi(U_\Theta) \subset M$, $\boldsymbol{A}^T\boldsymbol{B}$ is invertible.

Next, it is a well-known fact that $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a local diffeomorphism and thus there exists an open neighborhood $U_2 \subset M$ containing $[\boldsymbol{A}, s]$ such that the logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}$ is well-defined and is smooth in U_2.

We now compute $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}$ for $[\boldsymbol{B}, t] \in U := U_1 \cap U_2$. Suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ satisfies $\boldsymbol{v} = \mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t])$. By Proposition 2, $[\boldsymbol{B}, t] = [\boldsymbol{A} + \boldsymbol{V}, s + c]$, where $(\boldsymbol{V}, c)$ is the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{A}, s)$. It follows that $c = t - s$ and that there exists a $\boldsymbol{Q} \in O(q)$ such that $\boldsymbol{V} = \boldsymbol{BQ} - A$. Or equivalently:

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BQ} - \boldsymbol{A}, t - s)). \tag{1.13}$$

Given that $(\boldsymbol{V}, c)$ is a horizontal tangent vector, Proposition 1 shows that

$$\boldsymbol{A}^T(\boldsymbol{BQ} - \boldsymbol{A}) = (\boldsymbol{BQ} - \boldsymbol{A})^T\boldsymbol{A}$$

and consequently $\boldsymbol{A}^T\boldsymbol{BQ}$ is symmetric. Lemma 3 with $\boldsymbol{L} = \boldsymbol{A}^T\boldsymbol{B}$ shows the existence of $\boldsymbol{R}$, $\boldsymbol{P} \in O(q)$, a diagonal matrix $\boldsymbol{S}$ with the singular values of $\boldsymbol{A}^T\boldsymbol{B}$ in increasing order, and a $q \times q$ diagonal matrix $\boldsymbol{D}$ with diagonal entries 1 or -1 such that $\boldsymbol{A}^T\boldsymbol{B} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$. (1.13) now becomes

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BPDR}^T - \boldsymbol{A}, t - s)). \tag{1.14}$$

Now let $\boldsymbol{B} \to \boldsymbol{A}$ and $t \to s$. The left hand size of (1.14) now becomes $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{A},s]) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M$. The singular value decomposition converges to the orthogonal diagonalization because $\boldsymbol{A}^T\boldsymbol{A}$ is symmetric, meaning that $\boldsymbol{P} = \boldsymbol{R}$. Thus

$$\boldsymbol{A}\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T - \boldsymbol{A} = \boldsymbol{0}.$$

Therefore, $\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T = \boldsymbol{I}_q$ and $\boldsymbol{D} = \boldsymbol{I}_q$. Lemma 4 then shows that

$$\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{R}^T = (\boldsymbol{B}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A}.$$

This completes the proof. □

Remark 6. Unfortunately, the logarithmic map is not globally defined in the standard way even over all $[\boldsymbol{B},t]$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible. For a counterexample, take $p=2$, $q=1$, $\boldsymbol{A}=[1,0]^T$, $\boldsymbol{B}=[1,1]^T$ and $s=t=1$. In this case, $O(1)=\{1,-1\}$. Then, although $\boldsymbol{A}^T\boldsymbol{B}=1$ is invertible, we still have

$$\operatorname{Exp}_{[\boldsymbol{A},s]}\left(d\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}-\boldsymbol{A},0))\right) = [\boldsymbol{B},1] = [-\boldsymbol{B},1] = \operatorname{Exp}_{[\boldsymbol{A},s]}\left(d\pi_{(\boldsymbol{A},s)}((-\boldsymbol{B}-\boldsymbol{A},0))\right).$$

The reason is that, even when we assume $\boldsymbol{A}^T\boldsymbol{B}$ is invertible, deducing $\boldsymbol{D}=\boldsymbol{I}_q$ relies on the local smoothness of the logarithmic map and the procedure of letting $[\boldsymbol{B},t] \to [\boldsymbol{A},s]$. If $[\boldsymbol{B},t]$ is far away from $[\boldsymbol{A},s]$, then $\boldsymbol{Q}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$ with some $\boldsymbol{D}\neq\boldsymbol{I}_q$ can still produce a valid horizontal vector whose image under $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is $[\boldsymbol{B},t]$, so that $\operatorname{Exp}_{[\boldsymbol{A},s]}$ fails to be injective and admits no globally defined inverse.

Nevertheless, the choice $\boldsymbol{D}=\boldsymbol{I}_q$ has an interesting property of selecting the geodesic of minimal speed and length when $[\boldsymbol{B},t]$ is any point such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible. To see this, consider a general candidate $\boldsymbol{Q}'=\boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$ and write $\boldsymbol{v}'=d\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s))$, so that $(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s)$ is the horizontal lift of $\boldsymbol{v}'$ at $(\boldsymbol{A},s)$ and $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}')=[\boldsymbol{B},t]$. Let us compute the squared norm of $\boldsymbol{v}'$:

$$\begin{aligned} g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}') &= g^{\Theta}_{(\boldsymbol{A},s)}\left((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s),(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s)\right) \\ &= \operatorname{tr}\left((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A})(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A})^T\right)+(t-s)^2 = \operatorname{tr}\big(\boldsymbol{B}\boldsymbol{B}^T\big)+\operatorname{tr}\big(\boldsymbol{A}\boldsymbol{A}^T\big)-2\operatorname{tr}(\boldsymbol{S}\boldsymbol{D})+(t-s)^2, \end{aligned}$$

where the last equality substituted $\boldsymbol{A}^T\boldsymbol{B}\boldsymbol{Q}'=\boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T\boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T=\boldsymbol{R}\boldsymbol{S}\boldsymbol{D}\boldsymbol{R}^T$ and used the cyclic property of the trace. The fact that the diagonal entries of $\boldsymbol{S}$ are strictly positive show that $g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')$ is minimized precisely when $\boldsymbol{D}=\boldsymbol{I}_q$. Given that the geodesic starting at $[\boldsymbol{A},s]$ with initial velocity $\boldsymbol{v}'$ reaches $[\boldsymbol{B},t]$ at time 1 with constant speed $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')}$, its length is $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')}$. The choice $\boldsymbol{D}=\boldsymbol{I}_q$ gives the slowest and hence shortest geodesic. To illustrate, let us go back to the counterexample in the previous paragraph. Put $\boldsymbol{v}_\pm := d\pi_{(\boldsymbol{A},s)}((\pm\boldsymbol{B}-\boldsymbol{A},0))$. Direct computation shows $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_+,\boldsymbol{v}_+)=1$ and $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_-,\boldsymbol{v}_-)=5$ so that the option corresponding to $\boldsymbol{D}=\boldsymbol{I}_1=1$ gives the slowest speed. Therefore, it is actually possible to define $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}$ over all $[\boldsymbol{B},t]$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible by manually asking it to output the minimal speed candidate and this definition will yield the same $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ formula.

However, the situation totally breaks down in case $\boldsymbol{A}^T\boldsymbol{B}$ is not invertible. Consider again $p=2$, $q=1$, $\boldsymbol{A}=[1,0]^T$, $s=t=1$, but this time $\boldsymbol{B}=[0,1]^T$. Then $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_+,\boldsymbol{v}_+)=g_{[\boldsymbol{A},s]}(\boldsymbol{v}_-,\boldsymbol{v}_-)=2$ and there are two distinct geodesics of the same speed so $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t])$ cannot be well-defined.

(Picture: there is a hole at $[0,0]^T$. Starting $\boldsymbol{A}=[1,0]^T$, as $\boldsymbol{B}=[0,1]^T$ is glued to $[0,-1]^2$, $\boldsymbol{v}_+$ points from $[1,0]^T$ to $[0,1]^T$, $\boldsymbol{v}_-$ points from $[1,0]^T$ to $[0,-1]^T$)

Given the above situation and (the expectation that) our work should not require a global definition, we stick to the standard definition of logarithmic map only defined locally around $[\boldsymbol{A},s]$.

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.6) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in [Sun et al.(2026)Sun, Lin, and Liu], definition III.1 and III.2.

Let us introduce a few notations. For each $[\boldsymbol{A}, s] \in M$, let

$$p(\boldsymbol{x}; [\boldsymbol{A}, s]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\boldsymbol{\Sigma}([\boldsymbol{A}, s])) \exp\left(-\frac{1}{2}\boldsymbol{x}^T (\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1}\boldsymbol{x}\right) \tag{1.15}$$

be the density of $N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))$. Define two maps $l_{[\boldsymbol{A},s]}$ and $r_{[\boldsymbol{A},s]}$ with domain $\mathcal{E}_{[\boldsymbol{A},s]} \subset T_{[\boldsymbol{A},s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$ and codomain $L^2(\mathbb{R}^p)$ by setting, for all $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$,

$$l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \log\left(p(\boldsymbol{x}; \mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right) \tag{1.16}$$

and

$$r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \sqrt{p(\boldsymbol{x}; \mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))}. \tag{1.17}$$

Theorem 7. *$r_{[\boldsymbol{A},s]}$ is Fréchet differentiable at any $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score*

$$s(\boldsymbol{x}; [\boldsymbol{A}, s])(\boldsymbol{h}) = \mathrm{tr}\left\{(\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - (\boldsymbol{\Sigma}([\boldsymbol{A}, s]))\right](\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1}(D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{A}, s]))\right\}, \tag{1.18}$$

where

$$D_{\boldsymbol{h}}\boldsymbol{\Sigma}([\boldsymbol{A}, s]) := \left.\frac{d}{dt}\right|_{t=0} \boldsymbol{\Sigma}(Exp_{[\boldsymbol{A},s]}(t\boldsymbol{h})) = 2\boldsymbol{U}\boldsymbol{A}^T + b\boldsymbol{I}_p, \tag{1.19}$$

if $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ lifts to $(\boldsymbol{U}, b)$ at $(\boldsymbol{A}, s)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\boldsymbol{A},s]}(\boldsymbol{h}_1, \boldsymbol{h}_2) = \mathrm{tr}\left\{(\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1} D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}([\boldsymbol{A}, s])(\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1} D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}([\boldsymbol{A}, s])\right\} \tag{1.20}$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\boldsymbol{A},s]}$ at $\boldsymbol{v} \in \mathcal{E}_{[\boldsymbol{A},s]}$ in a given direction $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\boldsymbol{v} = \boldsymbol{0}$ to prove (1.18) and (1.20).

Given $\boldsymbol{v}$ and $\boldsymbol{h}$, there exists an $\epsilon > 0$ such that $\boldsymbol{v} + t\boldsymbol{h} \in \mathcal{E}_{[\boldsymbol{A},s]}$ whenever $t \in (-\epsilon, \epsilon)$. If they respectively lift to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ at $(\boldsymbol{A}, s)$, then $\boldsymbol{v} + t\boldsymbol{h}$ lifts to $(\boldsymbol{V} + t\boldsymbol{U}, c + tb)$ at $(\boldsymbol{A}, s)$. First of all:

$$\begin{aligned} D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})) &:= \left.\frac{d}{dt}\right|_{t=0} \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v} + t\boldsymbol{h})) = \left.\frac{d}{dt}\right|_{t=0}\left\{(\boldsymbol{A} + \boldsymbol{V} + t\boldsymbol{U})(\boldsymbol{A} + \boldsymbol{V} + t\boldsymbol{U})^T + (s + c + tb)\boldsymbol{I}_p\right\} \\ &= \boldsymbol{U}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{U}^T + 2\boldsymbol{U}\boldsymbol{A}^T + b\boldsymbol{I}_p, \end{aligned} \tag{1.21}$$

proving (1.19) with special case $\boldsymbol{v} = \mathbf{0}$. Secondly, for all $\boldsymbol{x} \in \mathbb{R}^p$:

$$\begin{aligned}
&D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) := \frac{d}{dt}\bigg|_{t=0} l_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x}) \\
&= \frac{d}{dt}\bigg|_{t=0} \left\{-\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}))\right) - \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}))\right)^{-1}\boldsymbol{x}\right\} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(\frac{d}{dt}\bigg|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\right] \\
&+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(\frac{d}{dt}\bigg|_{t=0}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\right] \\
&+ \frac{1}{2}\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x} \\
&= \frac{1}{2}\operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left[\boldsymbol{x}\boldsymbol{x}^T - \left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\right]\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\right\}.
\end{aligned} \tag{1.22}$$

Finally:

$$D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = D_{\boldsymbol{h}}\exp\left(\frac{1}{2}l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})\right) = \frac{1}{2}\exp\left(\frac{1}{2}l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})\right)D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \frac{1}{2}r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}). \tag{1.23}$$

To prove that (1.23) indeed defines a valid Gâteaux derivative, we must prove that $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in [Gupta and Nagar(2000)] For $\boldsymbol{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, Theorem 7.7.1(i) states that

$$\mathbb{E}\left[\boldsymbol{X}^T\boldsymbol{Z}\boldsymbol{X}\right] = \operatorname{tr}(\boldsymbol{Z}\boldsymbol{\Sigma})$$

for any $p \times p$ matrix $\boldsymbol{Z}$ and Theorem 7.7.2 states that

$$\operatorname{Cov}\left(\boldsymbol{X}^T\boldsymbol{Z}\boldsymbol{X}, \boldsymbol{X}^T\boldsymbol{Y}\boldsymbol{X}\right) = \operatorname{tr}\big(\boldsymbol{Z}\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big) + \operatorname{tr}\big(\boldsymbol{Z}^T\boldsymbol{\Sigma}\boldsymbol{Y}^T\boldsymbol{\Sigma}\big)$$

for any two $p \times p$ matrices $\boldsymbol{Y}$ and $\boldsymbol{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= -\frac{1}{2}\operatorname{tr}\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\right] \\
&+ \frac{1}{2}\int_{\mathbb{R}^p}\left[\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x}\right] p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = 0
\end{aligned}$$

and, whenever $\boldsymbol{h}_1$ and $\boldsymbol{h}_2 \in T_{[\boldsymbol{A},s]}M$:

$$\begin{aligned}
&\int_{\mathbb{R}^p} D_{\boldsymbol{h}_1} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) D_{\boldsymbol{h}_2} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= \int_{\mathbb{R}^p}\left[\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x}\right] \\
&\left[\boldsymbol{x}^T\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\left(D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1}\boldsymbol{x}\right] p(\boldsymbol{x}; \boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} \\
&= \operatorname{tr}\left\{\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_1}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}_2}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right\}.
\end{aligned} \tag{1.24}$$

Thus, in the particular case $\boldsymbol{h}=\boldsymbol{h}_1=\boldsymbol{h}_2$:

$$
\begin{aligned}
&\|D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|^{\frac{1}{2}} = \int_{\mathbb{R}^p} \left[D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})\right]^2 \mathrm{d}\boldsymbol{x} \\
&= \frac{1}{4}\int_{\mathbb{R}^p} \left(D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})\right)^2 p(\boldsymbol{x};\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v})))\mathrm{d}\boldsymbol{x} = \frac{1}{2}\,\mathrm{tr}\left\{\left[\left(\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right)^{-1} D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right]^2\right\}.
\end{aligned}
\tag{1.25}
$$

Therefore, $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}) \in L^2(\mathbb{R}^p)$ and (1.23) is indeed a valid Gâteaux derivative at $\boldsymbol{v}$ in the direction $\boldsymbol{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$
R(\boldsymbol{v},\boldsymbol{h})(\boldsymbol{x}) := r_{[\boldsymbol{A},s]}(\boldsymbol{v}+\boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) - D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})
$$

and prove that

$$
\lim_{\boldsymbol{h}\to\boldsymbol{0}} \frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v},\boldsymbol{h}); L^2(\mathbb{R}^p)\|^2 = 0.
$$

To start, the fundamental theorem of calculus shows that

$$
\begin{aligned}
&r_{[\boldsymbol{A},s]}(\boldsymbol{v}+\boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \int_0^1 \frac{d}{dt} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t \\
&= \int_0^1 \lim_{s\to 0}\frac{r_{[\boldsymbol{A},s]}(\boldsymbol{v}+(t+s)\boldsymbol{h})(\boldsymbol{x}) - r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})}{s}\mathrm{d}t = \int_0^1 D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x})\mathrm{d}t.
\end{aligned}
$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$
\begin{aligned}
&\|R(\boldsymbol{v},\boldsymbol{h}); L^2(\mathbb{R}^p)\| = \|\int_0^1 \left(D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h})(\boldsymbol{x}) - D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})\right)\mathrm{d}t; L^2(\mathbb{R}^p)\| \\
&\leqslant \int_0^1 \|D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t = |\boldsymbol{h}|\int_0^1 \|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t
\end{aligned}
$$

and

$$
\begin{aligned}
&\frac{1}{|\boldsymbol{h}|^2}\|R(\boldsymbol{v},\boldsymbol{h}); L^2(\mathbb{R}^p)\|^2 \leqslant \left\{\int_0^1 \|D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\frac{\boldsymbol{h}}{|\boldsymbol{h}|}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2 \\
&\leqslant \left\{\int_0^1 \sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t\right\}^2.
\end{aligned}
$$

The proof will be complete once we can show that

$$
\lim_{\boldsymbol{h}\to\boldsymbol{0}}\int_0^1 \sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|\mathrm{d}t = 0. \tag{1.26}
$$

We mus therefore examine the continuity of $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})$, which is easy to see if we revisit the formulas. (1.21) shows that $D_{\boldsymbol{h}}\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Thus, (1.23) shows that, for each fixed $\boldsymbol{x}\in\mathbb{R}^p$, $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x})$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$ and (1.25) shows that $\|D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|$ is smooth in both $\boldsymbol{h}$ and $\boldsymbol{v}$. Combine the two facts, we see that $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\boldsymbol{h}$ and in $\boldsymbol{v}$. The integrand inside (1.26) is thus bounded uniformly over all $t\in[0,1]$ and all $\boldsymbol{h}$ in a compact neighborhood of $\boldsymbol{0}$ and it suffices to show that, for every $t\in[0,1]$, that

$$
\lim_{\boldsymbol{h}\to\boldsymbol{0}}\sup_{|\boldsymbol{u}|=1}\|D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| = 0, \tag{1.27}
$$

thanks to the bounded convergence theorem. To prove (1.27), let $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$ be a $g_{[\boldsymbol{A},s]}$ orthonormal basis where $d = pq + 1 - \frac{q(q-1)}{2}$. Every unit vector $\boldsymbol{u}$ can be expressed as $\boldsymbol{u} = \sum_{i=1}^d u_i \boldsymbol{e}_i$ with $\sum_{i=1}^d u_i^2 = 1$. By linearity of $D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{v})$ in $\boldsymbol{h}$, we have

$$\begin{aligned}
&\|D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| \leqslant \sum_{i=1}^d |u_i| \|D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| \\
&\leqslant \sqrt{d} \max_{i=1,\ldots,d} \|D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|.
\end{aligned}$$

Taking supremum yields

$$\sup_{|\boldsymbol{u}|=1} \|D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{u}} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\| \leqslant \sqrt{d} \max_{i=1,\ldots,d} \|D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) - D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}); L^2(\mathbb{R}^p)\|.$$

Given that $D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v}+t\boldsymbol{h}) \to D_{\boldsymbol{e}_i} r_{[\boldsymbol{A},s]}(\boldsymbol{v})$ in $L^2(\mathbb{R}^p)$ as $\boldsymbol{h} \to \boldsymbol{0}$ for every i, every t, and every $\boldsymbol{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.21), (1.22), and (1.23) simplifies greatly when $\boldsymbol{v} = \boldsymbol{0}$:

$$\begin{aligned}
&D_{\boldsymbol{h}} r_{[\boldsymbol{A},s]}(\boldsymbol{0})(\boldsymbol{x}) = \frac{1}{2} r_{[\boldsymbol{A},s]}(\boldsymbol{0})(\boldsymbol{x}) D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{0})(\boldsymbol{x}) \\
&= \frac{1}{2}\sqrt{p(\boldsymbol{x};[\boldsymbol{A},s])}\, \mathrm{tr}\left\{ (\boldsymbol{\Sigma}([\boldsymbol{A},s]))^{-1} \left[\boldsymbol{x}\boldsymbol{x}^T - (\boldsymbol{\Sigma}([\boldsymbol{A},s]))\right] (\boldsymbol{\Sigma}([\boldsymbol{A},s]))^{-1} (D_{\boldsymbol{h}} \boldsymbol{\Sigma}([\boldsymbol{A},s])) \right\}.
\end{aligned}$$

The score $s(\boldsymbol{x};[\boldsymbol{A},s])(\boldsymbol{h})$ therefore equals to $D_{\boldsymbol{h}} l_{[\boldsymbol{A},s]}(\boldsymbol{0})(\boldsymbol{x})$ and is indeed given by (1.18). The Fisher information operator (1.20) is obtained by taking $\boldsymbol{v} = \boldsymbol{0}$ in (1.24). To show that $G_{[\boldsymbol{A},s]}$ is positive definite, suppose that $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ is such that $G_{[\boldsymbol{A},s]}(\boldsymbol{h},\boldsymbol{h}) = 0$. (1.20) shows that the Euclidean norm of the matrix $(\boldsymbol{\Sigma}([\boldsymbol{A},s]))^{-1} D_{\boldsymbol{h}} \boldsymbol{\Sigma}([\boldsymbol{A},s])$ must be zero. Therefore, $D_{\boldsymbol{h}} \boldsymbol{\Sigma}([\boldsymbol{A},s]) = \boldsymbol{0}$ and, by (1.19), it must be the case that $\boldsymbol{h} = \boldsymbol{0}$. □

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

We now move to the core of the subject.

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